OPERATION: PARADOX TILES

TEAM NAME: BURNT OUT BRAINS

TEAM MEMBERS

Waniya Suhail
Laiba Shahzad
Ayesha Farooqui

Table of Contents

1. Introduction
- 1.1 Scope
- 1.2 Type Conventions

2. Target System
- 2.1 Platform
- 2.2 Graphics

3. Development System
- 3.1 Software

4. Specification
- 4.1 Concept
- 4.2 Story
- 4.3 Game Structure
- 4.4 Players
- 4.5 Action
- 4.6 Game Flow
- 4.7 Win/Lose Condition
- 4.8 Graphics

5. Gameplay
- 5.1 World
- 5.2 Landscape
- 5.3 Object Types
- 5.4 Control
- 5.5 Physics & Tile Mechanics
- 5.6 Design Philosophy

6. Front End
- 6.1 Intro
- 6.2 Menus/Screens
- 6.3 Feedback System

7. Game Flowchart

8. Development Tools
- 8.1 Game Engine
- 8.2 IDE

9. Credits/References

1. Introduction

1.1 Scope

This document defines the design, gameplay mechanics, and structure of *Operation: Paradox Tiles*, a single-player first-person puzzle platformer focused on navigating a tile-based simulation with broken physics.

1.2 Type Conventions

- Player actions are described using control inputs (e.g., Arrow Keys, Space)
- Tile effects are described as triggered events
- UI messages are presented in quotes

2. Target System

2.1 Windows

Platform: PC (Windows)

2.2 Graphics

- 3D Graphics
- First-Person Perspective

3. Development System

3.1 Software

- Programming Language: C#

4. Specification

4.1 Concept

The player takes on the role of a spy trapped inside a mysterious simulation chamber. The room is made up of tiles, and each tile triggers a different effect when stepped on. Some tiles activate traps such as disappearing floors, gravity shifts or slowed down motion. The objective is to carefully navigate the room, learn tile behaviors and reach the exit before time runs out.

4.2 Story

“You have successfully retrieved the required documents. However, in order to leave you must navigate this hall and find the exit. The success of this mission lies on your shoulders.”

4.3 Game Structure

- One continuous level
- Difficulty increases progressively

4.4 Players

- Single-player only

4.5 Action

- Move across tiles
- Trigger tile effects
- Adapt to changing physics and controls
- Avoid hazards
- Reach exit

4.6 Game Flow

- Start screen
- Briefing
- Gameplay begins
- Player navigates hall
- Win: Mission Complete
- Lose: Timer runs out
- Replay

4.6 Win/Lose Condition

- Win: Reach the exit door before the timer expires
- Lose: TImer reaches zero

4.7 Graphics

- 3D environment
- First-person view

5. Gameplay

5.1 World

- Simulation chamber environment
- Tile-based structure

5.2 Landscape

- Hall with pillars and black and white tiles.

- Exit door which is the end goal.

5.3 Object Types

- Tiles (interactive triggers)
- Exit door (goal)
- Timer system (pressure mechanic)
- Hazards

5.4 Control

- Arrow Keys: Move
- Space: Jump
- Controls may dynamically change depending on tile effects

5.5 Physics

- Gravity inversion (player moves on ceiling)
- Time manipulation (speed up / slow down timer)
- Movement disruption (slow zones)
- Spatial displacement (teleportation)
- Perceptual illusion (distorted exit location)

5.6 Design Philosophy

The game explores “broken physics” as a core theme:

- Gravity behaves unpredictably
- Time is inconsistent
- Movement is unreliable
- Position changes happen spontaneously
- Things are not always as they seem

6. Front End

6.1 Intro

“Agent Paradox, you have successfully retrieved the required documents. However, in order to leave you must navigate this hall and find the exit. Good luck Agent! The success of this mission lies on your shoulders.”

6.2 Menus/Screens

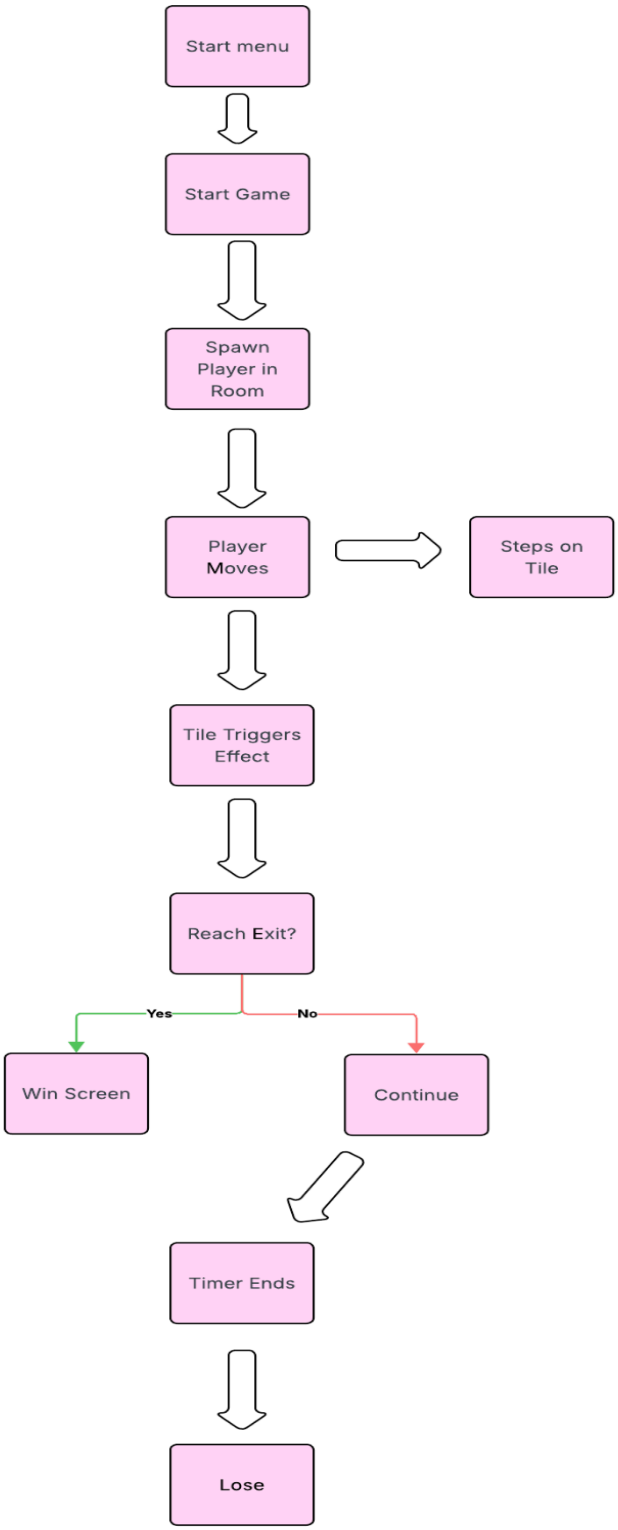
- Start Game
- Game Over
- You Win!

6.3 Feedback System

Each tile has virtual and text feedback

- “Gravity Inverted!”: screen flip
- “Time Warping!”: timer color change + pulse
- “Movement Disrupted!”: blue overlay
- “Displaced!”: glitch effect
- “Signal Lost... Relocating Exit”: distortion

7. Game Flowchart



8. Development Tools

8.1 Game Engine

Unity

8.2 IDE

Visual Studio

9. Credits/References

The following third-party assets and resources were used during the development of *Operation: Paradox Tiles*:

1. **3D Environment Model**

   abhayexe. (2025, July 11). VR Room [Light Baked] [3D Model]. Sketchfab. https://sketchfab.com/3d-models/vr-room-light-baked-4e4659da1542490dbfa3b3ca0d6bfd05

2. **Skybox / Environment Lighting**

   rpgwhitelock. (2024, July 20). AllSky Free - 10 Sky / Skybox Set [Unity Asset]. Unity Asset Store. https://assetstore.unity.com/packages/2d/textures-materials/sky/allsky-free-10-sky-skybox-set-146014

3. **UI Font**

   Jazz Create Games. (2025, December 22). Bubble Font (Free Version) [Unity Asset]. Unity Asset Store. https://assetstore.unity.com/packages/2d/fonts/bubble-font-free-version-24987

4. **Start Narration**

   Neosapience, Inc. (2026, April 17). Typecast (Margot) [AI voice generator]. https://typecast.ai/